

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

Annexure A

Experiments

Criteria	Understandin g of Concepts (2.5)	Experimental Design (3)	Use of Tools (2)	Analysis of Results (1.5)	Documentation (1)	Total(10)
Weightag e	25%	30%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments :

Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Age of child	Photograph:		
	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID Player Name Points Scored

P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16

P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.
(10 Marks)

Consider the Data Set

young,high,employed,good,yes
 young,high,self-employed,average,yes
 middle,high,employed,good,yes
 old,medium,employed,good,yes
 old,low,unemployed,poor,no
 old,low,self-employed,average,no
 middle,low,unemployed,poor,no
 young,medium,employed,average,yes
 young,low,unemployed,poor,no
 old,medium,self-employed,good,yes
 middle,medium,employed,average,yes
 middle,high,self-employed,good,yes
 old,medium,unemployed,poor,no
 young,medium,self-employed,average,yes
 middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 8476,35,47,64,95,66,89,36,84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 5051,56,84,60,59,70,63,66,50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

<i>age</i>	23	23	27	27	39	41	47	49	50
<i>%fat</i>	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
<i>age</i>	52	54	54	56	57	58	58	60	61
<i>%fat</i>	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

- (a) Calculate the mean, median, and standard deviation of *age* and *%fat*.
- (b) Draw the boxplots for *age* and *%fat*.
- (c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

Program File Edit Insert View Weka KnowledgeFlow Environment

Data mining processes Attribute summary Scatter plot matrix SQL Viewer Simple CLI

Design

Untitled1

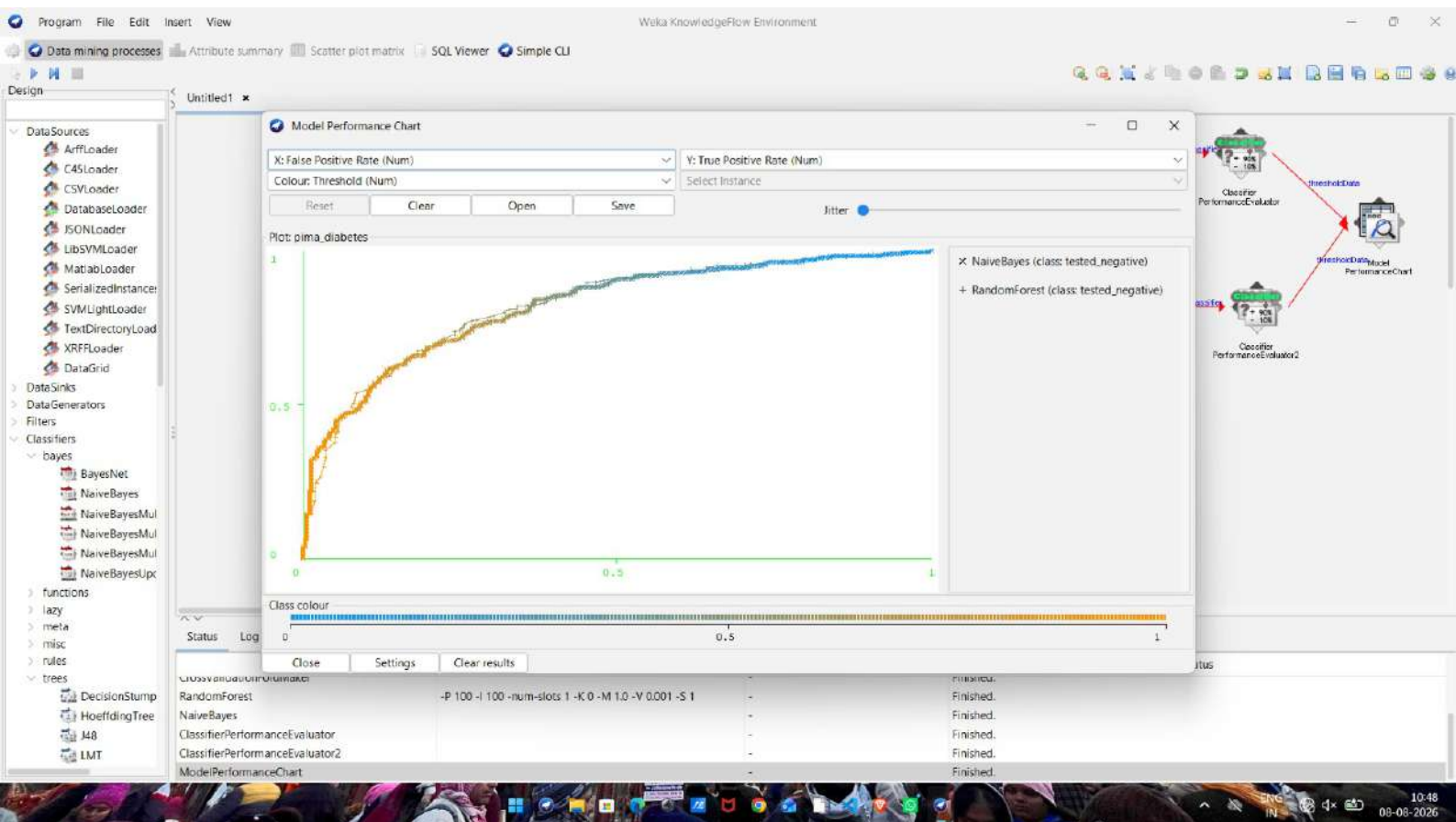
The workflow diagram in the Weka KnowledgeFlow Environment shows the following components and connections:

- ArffLoader** (Data Source) connects to **ClassAssigner** (Data Sink) via a **dataSet** connection.
- ClassAssigner** connects to **ClassValue Picker** (Data Sink) via a **dataSet** connection.
- ClassValue Picker** connects to **CrossValidation FoldSelector** (Data Sink) via a **dataSet** connection.
- CrossValidation FoldSelector** branches into two paths:
 - testSet** (trainingSet) connects to **NaiveBayes** (Classifier).
 - testSet** (trainingSet) connects to **RandomForest** (Classifier).
- NaiveBayes** connects to **Classifier Performance Evaluator** (Data Sink) via a **batchClassifier** connection.
- RandomForest** connects to **Classifier Performance Evaluator2** (Data Sink) via a **batchClassifier** connection.
- Classifier Performance Evaluator** connects to **ThresholdDataModel Performance Chart** (Data Sink) via a **thresholdData** connection.
- Classifier Performance Evaluator2** connects to **ThresholdDataModel Performance Chart** via a **thresholdDataModel** connection.

Status Log

Component	Parameters	Time	Status
CrossValidationFoldSelector			Finished.
RandomForest	-P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1		Finished.
NaiveBayes			Finished.
ClassifierPerformanceEvaluator			Finished.
ClassifierPerformanceEvaluator2			Finished.
ModelPerformanceChart			Finished.

10:47 09-09-2026



Sclab 2026.1.0 Console

File Edit Control Applications ?

File Browser

C:\Users\deepa\

Documents

Arduino

Custom Office Templates

IBM

My Tableau Repository

Project Backups

Rockstar Games

Sound Recordings

SQL Server Management Stud

Virtual Machines

Visual Studio 2017

.Rhistory

ADC Capstone ppt 1.pptx

ai ppt final.pptx

ai ppt.pptx

arduino.pptx

arduino.pptx

DEEPAK_LAPTOP

Assignment-4 lab CWS (1).doc

avi ss.docx

avinash CAPSTONE PROJECT

avinash capstone .docx

avinash's Presentation.pptx

BEEE SCLAB PDF.pdf

CAP-15.docx

capstone docs.docx

Capstone Project PPT Sample

Capstone project report mode

cc ppt.pdf

cloud computing ppt.pdf

Cloudflare_WARP_2026.3.85

Cloudflare_WARP_2026.4.13

Cloudflare_WARP_2026.6.82

co1_s81_deepak (1).pptx

co1_s81_deepak (1).pptx

co1_s81_deepak.pptx

co1_s82_deepak.pdf

co1_s83_deepak.pdf

File Directory Filter

File Edit Format Options Window Execute ?

EXP 15.sci (C:\Users\deepa\OneDrive\Desktop\BEEE LAB\EXP 15.sci) - SciNotes

EXP 15.sci

```
1 clear;
2 c10;
3 mode(0);
4 a=10;
5 H=s^4+2*s^3+3*s^2+4*s+5;
6 //H=s^5+7*s^4+6*s^3+42*s^2+5*s+56;
7 disp(H,'The given characteristics equation 1-G(s)H(s)=');
8 c=coeff(H);
9 len=length(c);
10 r=roots_r(H);
11 disp(r,'Rouths table=');
12 z=0;
13 for i=1:len
14 if (r(i,1)<0)
15 z=z+1;
16 end
17 end
18 if (z==1)
19 printf('From Rouths table, it is clear that the system is unstable. ');
20 else
21 printf('From Rouths table, it is clear that the system is stable. ');
22 end
23
```

Variable Browser

N...	Var...	T...	Vi...	M...
H	1x1 Pol...	local	288 B	
C	1x5 Do...	local	248 B	
i	5 Do...	local	216 B	
len	5 Do...	local	216 B	
r	5x3 Do...	local	328 B	
s	1x1 Pol...	local	264 B	
X	1 Do...	local	216 B	

Command History

-- 07/08/2026 14:03:12 --

News feed

Discover ...

Sclab is open source.

Giving back: support or use cases is appreciated.

- Add and improve functions on [SciLab](#).
- View and edit [help pages](#).

14:25 07-08-2026

Weka Experiment Environment

Setup Run **Analyse**

Source

Got 400 results

File... Database... Experiment

Actions

Perform test Save output Open Explorer...

Configure test

Testing with: Paired T-Tester (corrected)

Select rows and cols: Rows Cols Swap

Comparison field: Percent_Incorrect

Significance: 0.05

Sorting (asc) by: <default>

Test base: Select

Displayed Columns: Select

Show std. deviations: ☐

Output Format: Select

Test output

Testers: weka.experiment.PairedCorrectedTTester -G 4,5,6 -D 1 -R 2 -S 0.05 -result-matrix "weka.experiment.ResultMatrixPlainText -mean-prec 2 -stddev-prec 2 -c

Analysing: Percent_Incorrect

Datasets: 2

Resultsets: 2

Confidence: 0.05 (two tailed)

Sorted by: -

Date: 08/08/26, 10:28 am

Dataset: (1) bayes.Ra | (2) bayes

	(100)	24.75		24.25
pima_diabetes	(100)	28.44		50.55 v

(v/ /*) | (1/1/0)

Key:

(1) bayes.BayesNet '-D -Q bayes.net.search.local.X2 -- -P 1 -S BAYES -E bayes.net.estimate.SimpleEstimator -- -A 0.5' 746037443258775854

(2) bayes.NaiveBayes '' 5895281201785697655

Result list

10:21:31 - Available resultsets

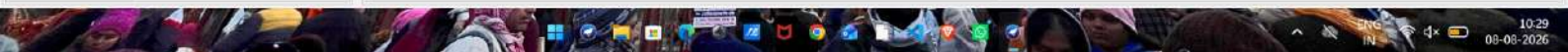
10:21:35 - Percent_correct - bayes.BayesNet '-D -Q bayes.net.se

10:21:44 - Percent_incorrect - bayes.BayesNet '-D -Q bayes.net

10:27:59 - Available resultsets

10:28:03 - Percent_correct - bayes.BayesNet '-D -Q bayes.net.se

10:28:13 - Percent_incorrect - bayes.BayesNet '-D -Q bayes.net



Weka Experiment Environment

SetupRunAnalyse

Source

Got 200 results

File...Database...Experiment

Actions

Perform testSave outputOpen Explorer...

Configure test

Testing with: Paired T-Tester (corrected)

Select rows and cols: RowsColsSwap

Comparison field: Percent_Incorrect

Significance: 0.05

Sorting (asc) by: <default>

Test base: Select

Displayed Columns: Select

Show std. deviations: ☐

Output Format: Select

Result list

10:21:31 - Available resultsets

10:21:35 - Percent_correct - bayes.BayesNet -D -Q bayes.net

10:21:44 - Percent_incorrect - bayes.BayesNet -D -Q bayes.net

Test output

Testers: weka.experiment.PairedCorrectedTTester -G 4,5,6 -D 1 -R 2 -S 0.05 -result-matrix "weka.experiment.ResultMatrixPlainText -mean-prec 2 -stddev-prec 2 -d

Analysing: Percent_Incorrect

Datasets: 1

Resultsets: 2

Confidence: 0.05 (two tailed)

Sorted by: -

Date: 08/08/26, 10:21 am

Dataset: (1) bayes.Ra | (2) bayes

pima_diabetes (100) 24.75 | 24.25

(v/ / ^) | (0/1/0)

Key:

(1) bayes.BayesNet '-D -Q bayes.net.search.local.X2 -- -P 1 -S BAYES -E bayes.net.estimate.SimpleEstimator -- -A 0.5' 746037443258775554

(2) bayes.NaiveBayes '' 5895281201785697655

